CALYX-LEVEL CARRIER rev A (Corallite Labs)
Plugs onto an Olimex ESP32-POE-ISO (WROOM) via EXT1/EXT2. One board for every role (sensor / pump / single):
6 level jacks (3.5 mm), high-level float that cuts pump power through a relay (no firmware in that path),
brushed pump on a MOSFET (pump B) or a TMC2209 stepstick (pump A). Notes: hardware/DESIGN.md, docs/schematic.md

OLIMEX ESP32-POE-ISO (WROOM)
EXT1/EXT2: 1x10, 2.54 mm, 25.4 mm apart
Carrier: female sockets. Olimex: male headers.
Boot straps: GPIO 0/2/5/12/15.
GPI39/GPI35: check the Olimex sense jumpers are open.

PWR_FLAG +5V
PWR_FLAG +3V3
PWR_FLAG GND

J1 EXT1
+5V
+3V3
GND
BUZZER
SPARE_IN2
HIGH_FLOAT_SENSE

1 +5V 2 +3V3 3 GND 4 EN 5 IO0
6 IO1/TX0 7 IO2 = BUZZER 8 IO3/RX0
9 IO4 = SPARE_IN2 10 IO5 = HIGH_FLOAT_SENSE

J2 EXT2
LVL_B
LVL_A
RO_LOW
SPARE_IN1
LEAK
PUMP_EN
STATUS_LED
DIR
STEP

1 GPI39 = LVL_B 2 GPI36 = LVL_A 3 GPI35 = RO_LOW
4 GPI34 (BUT1, on the Olimex) 5 IO33 = SPARE_IN1
6 IO32 = LEAK 7 IO16 = PUMP_EN 8 IO15 = STATUS_LED
9 IO14 = DIR 10 IO13 = STEP

LEVEL JACKS x6 (3.5 mm 4-pole; a 3-pole TRS plug works too: its sleeve lands on R2 + S)
T = signal, R1 = sensor supply (SENS_V through 100R, so a plug shorting during insertion can't pull the rail down),
R2 = GND through 0R (remove if an Apex plug uses R2), S = GND. VERIFY the Apex pinout before plugging Apex sensors in.
Default: 10k pull-up to 3V3 + 1k series + 100n (floats, open-collector or 3.3 V outputs; reads low = closed).
5 V push-pull sensor: R_PU DNP, R_SER 10k, R_DN 20k fitted -> 5 V reads 3.33 V.

J3 LVL_A
J4 LVL_B
J5 RO_LOW
J6 LEAK
J7 SPARE_IN1
J8 SPARE_IN2
S
R2
R1
T
SENS_V
L1_PWR
L1_R2
L1_TIP
L2_PWR
L2_R2
L2_TIP
L3_PWR
L3_R2
L3_TIP
L4_PWR
L4_R2
L4_TIP
L5_PWR
L5_R2
L5_TIP
L6_PWR
L6_R2
L6_TIP
100R R10
100R R20
100R R30
100R R40
100R R50
100R R60
DNP R14
DNP R24
DNP R34
DNP R44
DNP R54
DNP R64
10k R11
10k R21
10k R31
10k R41
10k R51
10k R61
+3V3
GND

PUMP SUPPLY 12 or 24 V (to match the pump; default build 24 V)
Barrel (PJ-102AH, 2.0 mm pin: Mean Well P1J plugs) or screw terminal, not both at once.
Fuse -> reverse-polarity Schottky -> TVS. The supply ground is the board GND: fine on the ISO board only.

L1_TIP
L2_TIP
L3_TIP
L4_TIP
L5_TIP
L6_TIP
PESD5V0S1BL
L1_TIP 10k R12
L2_TIP 10k R22
L3_TIP 10k R32
L4_TIP 10k R42
L5_TIP 10k R52
L6_TIP 10k R62
LVL_A
LVL_B
RO_LOW
LEAK
SPARE_IN1
SPARE_IN2
GND
LVL_A 10k R13
LVL_B 10k R23
RO_LOW 10k R33
LEAK 10k R43
SPARE_IN1 10k R53
SPARE_IN2 10k R63
C10 100nF
C20 100nF
C30 100nF
C40 100nF
C50 100nF
C60 100nF

SENSOR SUPPLY (SENS_V)
JP1 picks 3V3 or 5V
for all six jacks.
Default 3V3.

+3V3
SENS_V
+5V
JP1
C7 10uF

PUMP A (stepper, e.g. KPHM100/400-ST): TMC2209 stepstick, standalone.
VM = RAIL, so the high float also cuts the driver. VIO = 3V3.
EN (IO16) active low: R79 10k pull-up keeps it disabled at boot
and when the ESP32 is dead. MS1/MS2 jumpers to 3V3 (open = 1/8).
JP4: bridge only for A4988/DRV8825 (RST-SLP). J15 mates the Kamoer
XHP-4 lead: 1 RED, 2 YEL (coil B), 3 BRN, 4 ORG (coil A).
Don't plug or unplug the motor with RAIL live.

J13 STEPSTICK L
PUMP_EN
MS1
MS2
SS_RST
SS_SLP
STEP
DIR
J14 STEPSTICK R
RAIL
M_2B
M_2A
M_1A
M_1B
GND
+3V3
MS1 JP2
MS2 JP3
JP4 RST-SLP
R79 10k
C3 100uF 35V
C4 100nF 50V

STATUS LED (IO15) and BUZZER (IO2)
Both are boot straps: the LED and the base
pull-down hold them low at reset, which is
what the ESP32 wants (IO15 low only mutes
the ROM boot log).

STATUS_LED
R70 330R
D70 BLUE
BUZZER
R71 1k
BZ_B
R72 10k
Q70 MMBT3904
BZ_C
BZ1 5V active
D71 SW4148W
+5V
GND

PUMP SUPPLY 12 or 24 V (to match the pump: default build 24 V)
Barrel (PJ-102AH, 2.0 mm pin: Mean Well P1J plugs) or screw terminal, not both at once.
Fuse -> reverse-polarity Schottky -> TVS. The supply ground is the board GND: fine on the ISO board only.

J9 PUMP SUPPLY
VIN_RAW
F1 2A
VIN_F
SS34 D1
+VIN
SMAJ24A D2
C1 10uF 50V
C2 100nF 50V
R73 10k LED_VIN
D3 RED
J10 PUMP SUPPLY
GND

HIGH-LEVEL CUTOFF (hardware only). The float is closed while the water is below the high mark.
Float closed -> relay coil powered -> contact feeds RAIL. High water, a cut cable or a shorted cable
(F2 trips) drops the relay, and RAIL (pump + / stepstick VM) goes dead. Firmware only reads it (U4).
Cable: 2-core shielded pair, shield to J11-3 at this end only. 12 V build: G5LE-1 DC12, R74 2.2k.

+VIN
F2 200mA hold
FLOAT_V
FLOAT_RET
1N4148WS D4
K1 G5LE-1 DC24
RAIL
GND
FLOAT_K
R74 4.7k
D5 GREEN
+3V3
R75 10k
FOK_K
U4 PC817
HIGH_FLOAT_SENSE
J11 HIGH FLOAT
FLOAT_V
FLOAT_RET
GND

PUMP B (brushed, e.g. the 81 ml/min Kamoer): low-side MOSFET on IO16.
Runs only with RAIL live (float OK) AND the gate high. Dead ESP32 -> R77 holds the gate low.
Build B: R76 fitted, R79 DNP. Build A (stepper): R76 DNP (Q1 may be left off), R79 fitted.

RAIL
SS34 D6
PUMP_N
D7 YELLOW
R78 10k PUMP_PWR
PUMP_EN
R76 100R
GATE
Q1 IRLML0040
R77 10k
J12 PUMP
J15 MOTOR
M_2B
M_2A
M_1A
M_1B

TEST POINTS + MOUNTING
TP1 +VIN
TP2 RAIL
TP3 GATE
TP4 +5V
TP5 +3V3
TP6 GND
TP7 SENS_V
H1 M3
H2 M3
H3 M3
H4 M3

Generated by hardware/tools/gen_schematic.py; see hardware/DESIGN.md
Carrier for Olimex ESP32-POE-ISO: 6 level jacks, hardwired high-float cutoff, brushed or stepper ATO

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